

Beyond concordance

External validation of probabilistic grammar in the English dative alternation

June 2026

Abstract

This paper treats the external validation of a corpus-based probabilistic grammar as a full predictive-evaluation problem, with the English dative alternation as a worked case. Earlier studies have already trained dative models on one variety and applied them to another, reporting the transport as discrimination or classification accuracy. I freeze a production model fit to the languageR dative data of Bresnan et al. (2007) and evaluate its move to the later Spoken British National Corpus 2014 (BNC2014) conversational data with proper scoring, calibration, a paired target-native benchmark scored on the same observations, predictor-group ablation, missing-data sensitivity, and an explicit opportunity-set denominator. The source-trained profile retains substantial discrimination in the new corpus domain, and the portable part is mainly non-lexical conditioning rather than lexical base rates; but calibration, coefficient checks, and the paired native benchmark expose target-specific structure that discrimination alone hides. The fitted quantity is a conditional production probability among annotated alternation tokens, not a probability of grammaticality, so the validated model projects within a stated target domain while staying silent, by construction, about the grammatical-but-unattested region. The result is a validation ladder for probabilistic grammar and an account of what each rung licenses.

Keywords: dative alternation; probabilistic grammar; external validation; cross-corpus transport; calibration; projectibility; corpus linguistics

I THE PROBLEM

A corpus-based probabilistic grammar can predict held-out observations from its own source corpus and still fail when it's moved to another corpus. External validation needs more than a single discrimination statistic. It needs probability scoring, calibration, comparison with target-native predictions on the same observations, an account of which predictors carry the transport, and a specification of the opportunity set over which the predictions are defined. This paper assembles that regime and runs it on a canonical case. Throughout, PROBABILISTIC GRAMMAR names the fitted production model, not a claim about speakers' competence or a community's licensing of forms; external validation bears on the model's predictive transport, not directly on either.

The English dative alternation is that case. Speakers can say *give Kim the book* or *give the book to Kim*, and the choice is uneven: it varies with verb, length, animacy, definiteness, pronominality, discourse status, corpus, and register. The unevenness is easy to misread, because a rare pattern can

be lexically restricted, strongly dispreferred, or tied to a discourse context without falling outside the grammar. Bresnan et al. (2007) modelled that conditioning and argued that corpus probabilities can reveal grammatical possibilities informal judgements miss.

The model has since been shown to travel: trained on one variety and scored on another, it keeps much of its accuracy (Bresnan & Hay, 2008; Engel et al., 2025; Kendall et al., 2011). The open question isn't whether it travels but how well, once transport is treated as a full predictive-evaluation problem, and what a fitted production probability can and can't say about grammar. That last is a question of PROJECTIBILITY, what observing a profile in one setting licenses about another (Goodman, 1955).

The argument proceeds by separating validation claims that are often collapsed. Reproducing the source model asks whether the classic production profile is stable in its original data. Scoring frozen predictions in BNC2014 asks whether that profile carries predictive information out of domain. Comparing those predictions with a target-native model on the same rows asks how much local structure the source model misses. Calibration and ablation then ask whether the transport is a matter of ranking, probability level, lexical base rates, or non-lexical conditioning. Finally, an acceptability bridge asks where a production probability is licensed, and where it isn't, when carried into a judgement channel.

2 THE ORIGINAL TARGET

Bresnan et al. (2007) is the anchor because it treats the alternation as a grammatical problem, not a simple frequency count: the claim isn't that whatever occurs often is grammatical, but that observed choices expose systematic constraints on what speakers treat as available. That matters because informal judgements compress several facts into one verdict. A construction may sound odd because it's impossible, because it's rare, because the wrong verb is present, because a discourse condition is missing, or because the constructed example strips away the licensing context. Corpus data can help separate these cases.

The empirical target is production choice. In Bresnan et al.'s data the response is whether the recipient appears as a noun phrase (NP), in the double-object pattern, or as a prepositional phrase (PP), in the prepositional dative, with predictors for the verb, recipient, theme, discourse status, and corpus. This paper preserves that target and treats the production probability as evidence to interpret, not as a direct measure of grammaticality.

3 DATA REPRODUCTION AND REANALYSIS

The analysis starts with `languageR::dative`, distributed with the Comprehensive R Archive Network (CRAN) `languageR` package (Baayen & Shafaei-Bajestan, 2025). The source tarball supplies the packaged `dative.rda` object; the analysis scripts load it and write derived summaries.

The package documentation identifies the sources as Switchboard and the Treebank Wall Street Journal collection. Those sources place the original data in American English: Switchboard is telephone conversation among speakers from the United States, developed in 1990–1991 (Godfrey & Holliman, 1993), and the Treebank Wall Street Journal component is late-1980s American news and financial writing (Charniak et al., 2000; Marcus et al., 1999).

In the packaged dataset, there are 3,263 rows and 15 columns. The response records recipient realization, with 2,414 NP realizations and 849 PP realizations. The dataset contains 75 verb levels

and two modalities: 2,360 spoken tokens and 903 written tokens. Speaker identifiers are missing exactly for the written rows.

For the first model sequence, the response stays binary: NP is coded as 1 and PP as 0. The initial split is verb-stratified, with a fixed seed (20260623), so every verb observed in the test set is also represented in training. That split gives 2,637 training rows and 626 test rows.

The denominator is narrow. These rows are already tokens in which a dative recipient was realized as NP or PP. They aren't all contexts in which a transfer event could have been expressed, omitted, paraphrased, or packaged in some other construction. The models estimate conditional choice within an attested alternation frame: given that a token enters the dative-alternation sample, which realization appears?

Initial analyses are deliberately plain. A marginal model sets the baseline. Structured models then add modality, length, verb, and the non-verb predictors. Negative-control and simulation checks accompany them: a scrambled-label check and fake-data checks. These tests ask whether the fitted structure beats chance, not whether it proves a grammatical theory.

A second empirical source is the Spoken British National Corpus 2014 (BNC2014) dative dataset (G. Jensen & McGillivray, 2018; G. B. Jensen & McGillivray, 2019). The verified Figshare comma-separated values (CSV) file has 1,839 parsed data rows and 44 columns under a Creative Commons Attribution 4.0 International (CC BY 4.0) licence.

The BNC2014 dative file comes from the Early-Access Subset of the Spoken BNC2014, whose spontaneous conversations were recorded in 2012–2014 and contain speaker and context metadata.

BNC2014 serves as a transport target rather than a second copy of `languageR::dative`. Its outcome field is `Pattern`, where the verb–noun–noun code `VNN` maps to NP realization and the verb–noun–prepositional-phrase code `VNPP` maps to PP realization. The six shared verbs are *give*, *send*, *show*, *sell*, *offer*, and *lend*. All six occur in the packaged `languageR` dataset.

4 MODERN MODELLING UPDATE

Does modern modelling change Bresnan et al.'s linguistic conclusion, or only repackage it statistically? Here, only the statistical packaging changes. The table reports log loss, area under the receiver operating characteristic curve (AUC), and a count of fixed-effect terms.

Two scores carry the paper, so it's worth fixing what they measure. Log loss is a proper scoring rule: it rewards probabilities that are both confident and correct, penalizes confident mistakes, is lower-is-better, and reaches zero only for perfect probabilistic prediction. AUC is the probability that a randomly chosen NP token is scored above a randomly chosen PP token; 0.5 is chance and 1 is perfect ranking. The two see different things. AUC sees only the ranking, whereas log loss is also sensitive to calibration, whether the predicted probabilities match observed rates, which is why both are reported.

Three models share a non-verb core of modality, semantic class, recipient length, recipient animacy, recipient definiteness, recipient pronominality, theme length, theme animacy, theme definiteness, theme pronominality, and recipient/theme accessibility. The fixed-verb model adds verb as a fixed effect. The hierarchical model uses the same non-verb fixed effects and a varying verb intercept, with partial pooling pulling each verb's estimate toward the overall rate so that rare verbs borrow strength rather than over-fit. All are logistic models for NP versus PP realization with R treatment contrasts and no interactions.

Model	Test log loss	Test AUC	Fixed terms
Marginal null	0.554	0.500	1
Non-verb main model	0.271	0.932	18
Fixed-verb full model	0.234	0.951	92
Hierarchical verb-intercept model	0.233	0.952	18

Table 1: Held-out languageR: :dative results for the main model sequence.

The non-verb model already predicts held-out production choices well. Adding fixed verb effects improves test log loss from 0.271 to 0.234 and raises AUC from 0.932 to 0.951. The improvement confirms that lexical conditioning isn't a minor residue after length, animacy, definiteness, pronominality, and accessibility are included.

Partial pooling reaches almost the same held-out performance for far less: the hierarchical model's test log loss is 0.233 and its AUC is 0.952. It carries 18 fixed-effect terms plus a single verb-intercept variance, not a separate coefficient for each of the 75 verbs; the estimated verb random-intercept standard deviation is 2.106, so the lexical signal remains large under partial pooling.

Negative-control and simulation checks behave as they should. Scrambled-label fits lose predictive value. Fake-data fits without a verb signal collapse toward singular zero-variance random-effects models, while a fake verb-effect check recovers a nonzero verb standard deviation. These checks make the production result more credible, but they don't change its explanatory level.

The result supports Bresnan et al.'s production-choice conclusion. It doesn't create a new grammatical conclusion on its own. The same-data analysis shows that the classic profile is reproducible and statistically stable. The paper's new empirical pressure has to come from transport.

5 CROSS-CORPUS TRANSPORT

The BNC2014 analysis asks whether the production profile travels across corpus design, variety, period, register, and annotation regime. The transport model is trained on the spoken languageR rows restricted to the six BNC2014 verbs. It's then scored on harmonized BNC2014 rows, with complete-case filtering stated as part of the estimand, the quantity the analysis is actually trying to estimate.

Two corpus targets are kept apart. The languageR models estimate conditional production choice among attested NP and PP dative tokens. The BNC2014 analysis estimates out-of-domain predictive performance over complete-case rows in the six shared high-frequency verbs after harmonization. These are different evidence channels, not repeated measurements of one quantity. A third channel, controlled acceptability preference, is a different target again and is kept to the supplementary Dative Alternation and Information Structure (DAIS) bridge.

Out-of-domain transport of a dative model isn't new ground either. Bresnan and Hay (2008) extended the Bresnan et al. (2007) analysis to spoken New Zealand English; Kendall et al. (2011) trained a model on the Bresnan et al. (2007) data and predicted the alternation in African American English, then reversed the direction, reporting concordance indices of 0.97 and 0.95; and Engel et al. (2025) trained separate English and Dutch models and tested each on the other language. Bresnan and Ford (2010) and Röthlisberger et al. (2017) document broad cross-variety stability by other means. So neither the finding that dative conditioning travels nor the act of scoring one variety's model on an-

other is the contribution here.

What those transport studies report is discrimination: a concordance index, equivalently the area under the curve, or classification accuracy: whether the moved model ranks or labels new tokens correctly.

They don't ask whether it stays calibrated, how it compares to a target-native model fit and scored on the same observations, which predictor groups carry the transport, how missing data and the harmonized denominator bound the claim, or what quantity the fitted probability estimates. This paper treats external validation as that fuller problem. The contribution is the validation regime: proper scoring, calibration intercept and slope, a paired target-native benchmark on identical rows, predictor-group ablation, missing-data sensitivity, and an explicit opportunity-set denominator, with an open harmonization record that lets each step be checked.

A bundled out-of-domain test frames the comparison. The source side is American telephone conversation from 1990–1991, restricted to the six shared spoken languageR verbs. The target side is spontaneous British conversation from 2012–2014, mostly intimate face-to-face conversation, under a different annotation and extraction scheme. Those six verbs account for 2,201 of 3,263 languageR rows and for 1,564 of 2,360 spoken rows. They're the high-frequency core, not a random sample of the alternation.

The transport models are logistic generalized linear models with logit link. The core formula is

$$\begin{aligned} \text{Pr}(\text{NP}) \sim & \text{Verb} + \text{recipient length} + \text{theme length} \\ & + \text{recipient animacy} + \text{recipient pronominality} \\ & + \text{theme animacy} + \text{theme definiteness} + \text{theme pronominality}. \end{aligned}$$

Lengths are raw word counts, with no centering, scaling, or logging. Factors use R treatment contrasts, with *give* as the verb reference and “no” as the binary reference. There are no interactions in the transport models.

Factor levels and coding are fixed before scoring BNC2014; no BNC2014-specific scaling or imputation enters the headline model. Recipient definiteness is unavailable in BNC2014 and enters only as a labelled regex proxy in a sensitivity check. The validation is retrospective: freezing means the source coefficients aren't re-estimated from BNC2014 outcomes, not that the target corpus was unavailable while the harmonization, common-predictor set, and six-verb restriction were chosen. Those design choices are target-aware by necessity; the fitted coefficients aren't.

Harmonization is lossy. The outcome maps cleanly at the pattern level: VNN becomes NP and VNPP becomes PP. Verb maps cleanly after the six-verb restriction. Length needs more care. The released BNC2014 length fields are character counts: the data dictionary defines RecLen and ThemeLen as the number of characters in the recipient and theme (G. B. Jensen & McGillivray, 2019). Those fields aren't comparable to languageR's token-like lengths, so the scripts derive word counts from the released phrase strings. Rows with empty phrase strings can't be used in the complete-case transport model. The result is a harmonized analysis, not an assumption that the two extraction protocols are interchangeable.

Composition differs between target and source. In the six-verb spoken source rows, *give* accounts for 0.808 of tokens and *send* for 0.089; in the BNC2014 complete cases the corresponding shares are 0.450 and 0.300. Recipients are still usually pronominal in both corpora (0.831 in the source, 0.870 in BNC2014 complete cases), but recipient animacy is lower in the target (0.943 vs. 0.837),

and themes are shorter on average (3.68 vs. 2.43 words). These aren't nuisance details; they define the population shift against which transport is being tested.

The core transport model scores the 1,621 BNC2014 rows complete for the harmonized predictors, not all 1,839 released dative rows. The missing rows are outcome-skewed: NP recipients make up 0.819 of the complete rows but only 0.670 of the 218 incomplete rows. The variable-level pattern is also uneven. Recipient-pronominality is missing for 112 rows with NP rate 0.357, while theme-pronominality is missing for 139 rows with NP rate 0.950. The headline comparison is a complete-case estimate over the shared high-frequency core.

A BNC2014 metadata audit makes the domain concrete: the released dative rows are predominantly UK English (1,720 of 1,839), skew female (1,138) and young (the 19–29 band is the largest), and are dominated by close-family, partner, and very-close-friend conversations (1,171 rows). The complete-case subset keeps the same shape, so a single BNC2014 transport score is a corpus-domain diagnostic, not a clean estimate for British spoken English in general.

The released BNC2014 file also limits the resampling design. The supporting scripts show that an intermediate `main_speaker_id` was derived and selected, but it was dropped before the public 44-column CSV. No stable speaker, text, or conversation identifier is recoverable from the released files. The native BNC2014 benchmark uses repeated verb-stratified row-level cross-validation, not grouped speaker or conversation folds. The comparison below repairs the old sample mismatch by scoring source-trained and native out-of-fold (OOF) predictions, each row scored by a model trained on the other folds, on the same 1,621 rows, and by computing the AUC gap fold by fold; it can't remove within-conversation dependence.

Table 2 asks three questions on the same rows: whether the source model beats a target marginal baseline, whether a BNC2014-native model does better when trained on the target, and whether the transported signal comes from verb identity or from the non-lexical predictors.

Origin	Model family	Log loss	AUC
languageR source	marginal	0.473	0.500
languageR source	verb only	0.456	0.646
languageR source	non-verb	0.296	0.899
languageR source	full core	0.308	0.906
BNC2014-native OOF	marginal	0.473	0.500
BNC2014-native OOF	verb only	0.439	0.625
BNC2014-native OOF	non-verb	0.234	0.928
BNC2014-native OOF	full core	0.227	0.939

Table 2: Same-row BNC2014 complete-case evaluation. OOF = out-of-fold. All rows are the same 1,621 BNC2014 complete-case observations; the native rows use five repeated 10-fold verb-stratified row-level cross-validation. Native AUC is averaged within each held-out fold, with the source prediction scored on the same held-out fold for the AUC gap; log loss is pooled. The released file carries no speaker or conversation identifier, so the folds are row-level and within-conversation dependence isn't removed.

The source-trained non-lexical profile transports, but not because shared verb base rates do most of the work. The source full model improves log loss over the source marginal model by 0.165 in

a paired row bootstrap ($[-0.198, -0.133]$ for full minus marginal) and raises AUC by 0.406. The source verb-only model gives only a small log loss gain over marginal (0.456 vs. 0.473), while the source non-verb model reaches 0.296 log loss and 0.899 AUC. Adding the source verb terms to the non-verb model improves ranking slightly but worsens log loss slightly (full minus non-verb = 0.012, bootstrap interval $[-0.006, 0.030]$).

In this target corpus, the portable part of the profile is mainly the non-lexical conditioning, captured by the non-verb model, rather than lexical base-rate ordering.

Coefficient comparison gives a second, more local check. Fitting the same full formula separately to the source core and the BNC2014 complete cases gives the same sign for ten of twelve non-intercept terms. Recipient length is negative in both models, theme length positive, recipient pronominality strongly positive, and theme pronominality strongly negative. The two unstable signs are recipient animacy and theme definiteness, and the lexical comparison is mixed: the source *offer* penalty largely disappears in BNC2014. The transport claim isn't that every coefficient is invariant. It's that the length and pronominality profile, and much of the animacy profile, remain predictively useful under a substantial corpus shift.

Target-native performance remains better. On the identical rows, the BNC2014 native full model has log loss 0.227 and a fold-averaged AUC of 0.939. Against the source full model scored on the same held-out folds, its log loss is lower by 0.081 (paired row bootstrap $[-0.104, -0.059]$) and its fold-averaged AUC is higher by 0.032 (fold bootstrap $[0.025, 0.039]$); the source fold-averaged AUC in that matched comparison is 0.906. The native non-verb model is already close to the native full model, which reinforces the same interpretation: the broad conditioning profile is portable, but the target corpus supplies local calibration, annotation-regime adaptation, and residual structure.

Miscalibration also appears verb by verb: the source full model underpredicts NP realization for *send*, *offer*, and *lend*, and overpredicts it for *sell*. Apparent recalibration on the same BNC2014 rows, a diagnostic upper bound rather than out-of-sample performance, gives a calibration intercept of 0.44 (standard error 0.10) under an intercept-only update and an intercept of 0.58 (0.09) with slope 0.73 (0.04) under an intercept-and-slope update, lowering log loss from 0.308 to 0.301 and to 0.290 respectively. Even that best case stays short of the target-native full model's 0.227, so the remaining gap isn't removed by globally rescaling the level and slope of the source predictions; it isn't only a prevalence shift.

Across the probability range the miscalibration is easier to see than verb by verb. Figure 2 bins predictions against observed rates: the transported source model departs from the diagonal while the target-native model tracks it. The two models discriminate similarly but are calibrated differently, which is the property a concordance index can't show.

A matched check separates the two things the missing rows change: the predictor set and the row denominator. Only three predictors are observed for every BNC2014 row, so the reduced common-predictor model keeps verb, theme animacy, and theme definiteness. A model restricted to those predictors no longer recovers the transport result: on the 1,621 complete-case rows it scores 0.562 log loss and 0.510 AUC, worse than the source marginal baseline's 0.473 (Table 2) and far short of the full model's 0.308. Expanding that same reduced model from the complete-case rows to all 1,839 released rows changes performance much less (log loss 0.562 to 0.586, AUC 0.510 to 0.523) than the predictor reduction itself.

The complete-case claim depends principally on the availability of the recipient and length pre-

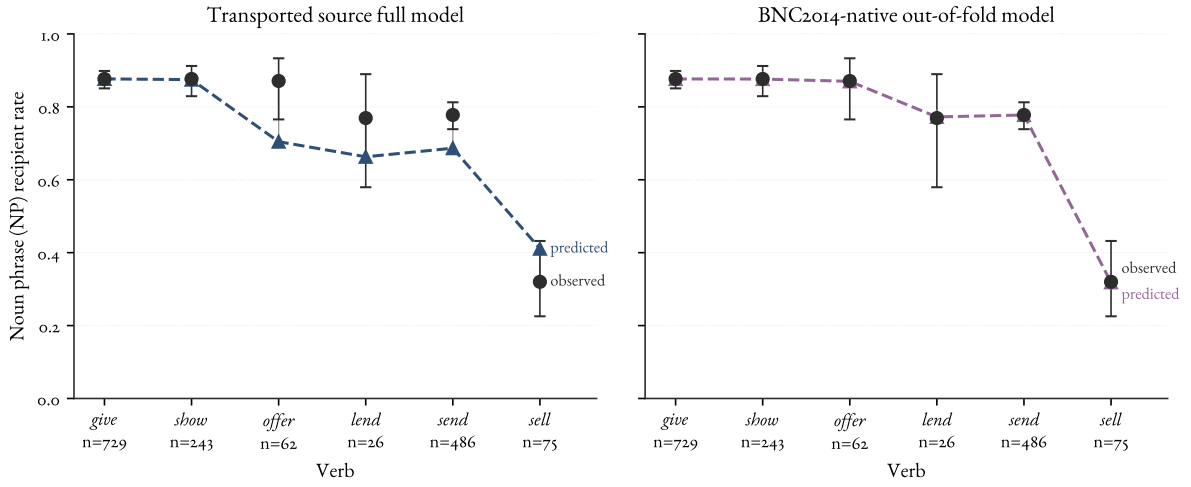


Figure 1: Verb-level calibration for the transported source full model and the BNC2014-native out-of-fold model on the same complete-case rows. Black points and intervals are observed NP recipient rates with binomial Wilson intervals, unadjusted for conversational clustering; coloured triangles are mean predictions.

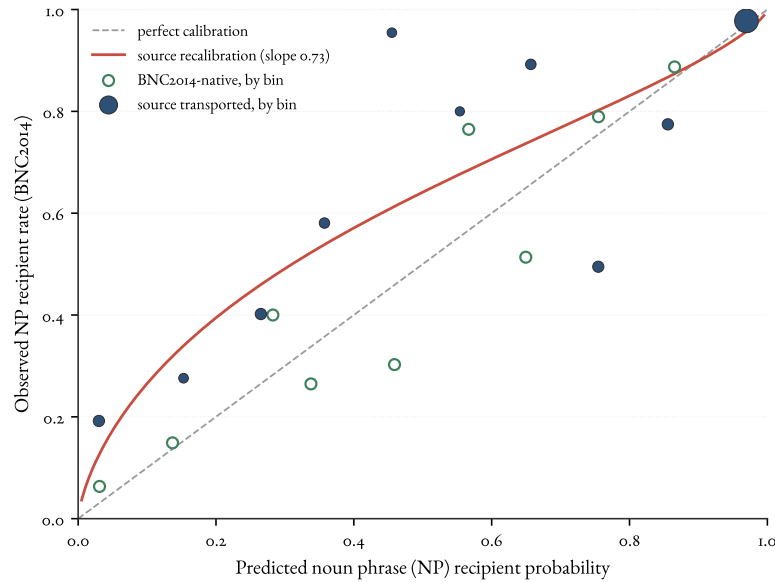


Figure 2: Reliability diagram on the BNC2014 complete-case rows. Points are binned observed noun phrase (NP) recipient rates against mean predicted probability, with marker size proportional to bin count. The transported source model departs from the diagonal (its fitted intercept-and-slope recalibration has slope 0.73), while the BNC2014-native model sits closer to it. Discrimination and calibration are different properties.

dictors, not on a favourable comparison between 1,621 and 1,839 rows.

6 ESTIMAND, OPPORTUNITY SET, AND INFERENCE SCOPE

The transport result is informative only against a clear statement of what the model estimates. The fitted probability is, approximately,

$$P(\text{NP recipient} \mid \text{a token entered the annotated NP/PP dative sample}),$$

not the probability that a double-object form is grammatical, and not the probability that a transfer event is realized as a dative at all. The corpus rows are already tokens in which a recipient was realized as an NP or a PP. They aren't the set of contexts in which a transfer event could have been expressed, paraphrased, left implicit, or judged. The models estimate conditional choice inside an attested alternation frame.

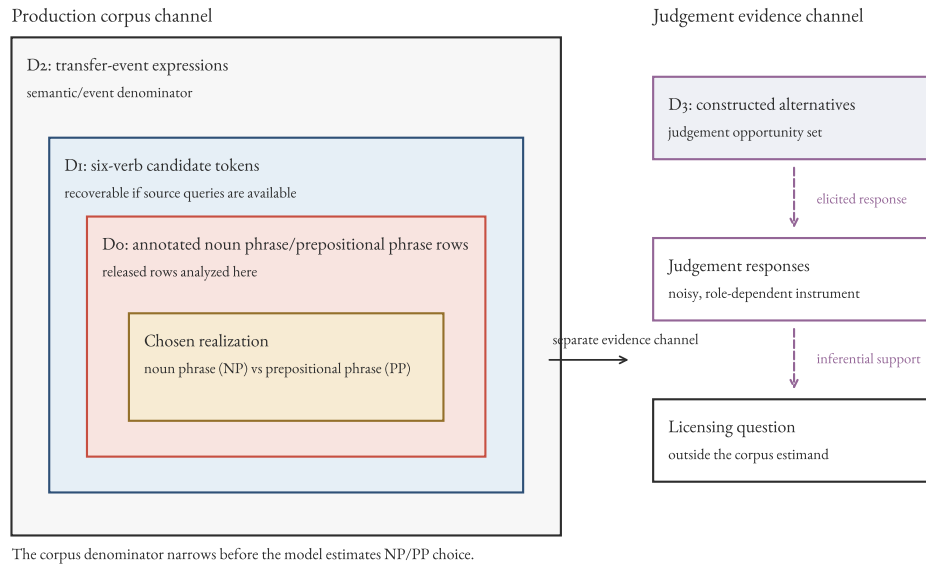


Figure 3: Nested denominators for the dative estimand. The released corpus rows fix realization choice inside the annotated noun phrase/prepositional phrase sample, the innermost set. The broader transfer-event and judgement sets are different denominators that answer different questions; reaching them is an inferential step to a new target, not a larger sample of the same one.

This bounds what successful transport licenses, and it separates two notions that sit close. The conditioning relations the profile encodes, the effects of pronominality, length, animacy, and definiteness, are projectible in Goodman's sense (Goodman, 1955): the kind of generalization that supports projection to new cases. Whether a fitted model then transports to a given new corpus is a separate, empirical question, and a projectible relation can still fail to transport under a population shift.

Within its demonstrated scope, the production model forecasts realization for unseen tokens among the verbs, registers, and varieties it covers. It stays silent, by construction, about grammatical but unattested cases. Additional positive tokens collected under the same sampling design D_0

can't identify that region by themselves; more tokens enlarge the attested NP/PP area without changing the opportunity set. That's an observational-identification claim, not a verdict that production evidence is useless. A broader denominator, elicited production, novel-item productivity tests, or acceptability judgements can each add information about the otherwise empty cells.

The acceptability channel illustrates the last route and its limits. A model validated for production choice isn't thereby validated for judgement. The supplementary DAIS bridge shows moderate marginal rank correspondence, a general DAIS shift toward the double-object option, and little incremental production-score contribution once the manipulated recipient, theme, and verb conditions are represented directly. The clearest verb-level residuals go in opposite directions: *offer* is more double-object-favouring in DAIS than its production score predicts, while *sell* is less so. Validity is relative to a target; carrying it across channels is a further claim, not an automatic one.

External validation is itself a ladder, but the useful entailment is failure localization rather than ascent: each rung marks a place where a transported model can pass or slip. The ladder borrows from prediction-model validation, especially the separation of discrimination, calibration-in-the-large, and calibration slope (Van Calster et al., 2016), and from the variationist insistence that a variable be analysed inside a defined envelope of variation (Labov, 1972; Weinreich et al., 1968). Table 3 lays out the rungs, and the dative shows what each reports. Source-data reproduction, within-domain validation, frozen cross-domain prediction, a target-native benchmark scored on the same observations, recalibration, and error decomposition are distinct steps, and a model can clear some while failing others.

Here the source model discriminates well out of domain but stays worse-calibrated than a target-native fit, and an intercept-and-slope recalibration doesn't close the gap, so the remaining gap isn't removed by globally rescaling the predictions; it isn't only a prevalence shift. Reporting only discrimination, as a concordance index or an accuracy, would have hidden that.

Stage	Data	Diagnostic	Licensed conclusion
Source reproduction	Source corpus	Held-out log loss and AUC against nulls	The source profile is real and reproducible
Within-source validation	Source corpus	Partial pooling, negative controls	Lexical and non-lexical signals aren't overfit
Frozen external scoring	Target corpus	Log loss and AUC against the target marginal	The profile carries predictive information out of domain
External calibration	Target corpus	Calibration intercept and slope	Whether the levels are right, not just the ranking
Paired target-native benchmark	Target corpus	Same-row paired loss differences	How much local structure the source model misses
Predictor-group ablation	Both	Marginal, verb-only, non-verb, and full contrasts	Which part of the profile transports
Opportunity-set audit	Target corpus	Denominator and missingness map	What quantity the predictions estimate

Table 3: A validation ladder for a corpus-based probabilistic grammar. Each stage has its own diagnostic and licenses a different conclusion, and a model can pass some rungs while failing others.

The ladder is a proposal for other alternations under stated preconditions: two corpora sharing a harmonizable categorical alternation, an annotated outcome, and a lexical core large enough to fit and to score on. Its limits carry over too. The comparison is testable only where the corpora intersect, so the periphery of an alternation goes unevaluated; preprocessing learned on the source has to be frozen before the target is scored; and the released data set the ceiling on how honestly dependence can be handled.

The Spoken BNC2014 release makes the last point concrete: it omits the speaker and conversation identifiers that grouped resampling needs, so the native benchmark can only use row-level cross-validation and can't remove within-conversation dependence.

That ceiling is a property of the data release, not a defect of the method. The row bootstrap and Wilson intervals reported above are conditional row-level intervals: they reflect test-row resampling around the fitted predictions, not source-model estimation uncertainty or within-conversation dependence, which probably makes them too narrow.

So the analyses license a cautious, well-scoped claim. The Bresnan et al. production-conditioning profile is durable and partly transportable within the shared high-frequency core; its non-lexical component travels better than its lexical base rates; and a target-native model still recovers local structure the source model misses. What the validated model buys is prediction within a stated domain, plus a clear marker of the low-probability cells where grammaticality, if it's at issue, has to be settled through another channel and another paper.

7 CONCLUSION

External validation of a probabilistic grammar is a ladder, and the dative shows what each rung reports. The first rung holds: the languageR profile survives baselines, negative controls, simulation checks, and partial pooling, and the hierarchical model matches the fixed-verb model with a smaller fixed-effect parameterization. Modern modelling confirms the classic production-choice conclusion rather than revising it.

The profile then partly transports. On the BNC2014 complete-case rows, the spoken six-verb languageR model beats a marginal baseline when it's scored against a target-native out-of-fold benchmark on the same observations. The portable signal is mainly non-lexical rather than verb base rates: the source non-verb model nearly matches the source full model. But the target-native model is better calibrated, and an intercept-and-slope recalibration doesn't close the gap, which a global rescaling of the source predictions would have done were it only a prevalence shift.

The coefficient comparison points the same way while exposing limits: length and pronominality travel more cleanly than every verb effect or definiteness proxy. The verdict is bounded. Transport survives the null comparisons in the high-frequency core, while variety, period, register, annotation, extraction protocol, and missingness stay bundled.

That contrast is visible only because the evaluation goes beyond concordance. Earlier transport studies report discrimination or accuracy; proper scoring, calibration, a native benchmark on the same rows, predictor-group ablation, and missing-data sensitivity distinguish a model that ranks new tokens from one calibrated to them. The regime, not the finding that the dative travels, is the contribution. Its forward payoff is a template for other alternations under the preconditions set out above, not a demonstration that all alternations behave the same way.

The estimand keeps the claim honest. The fitted probability is a conditional production choice

inside an annotated alternation sample, not a probability of grammaticality. A validated production model forecasts realization within a stated target domain and stays silent, by construction, about the grammatical-but-unattested region; no quantity of further positive tokens under the same design reaches it. Crossing that boundary needs a broader opportunity set or another evidence channel. The DAIS supplement gives one compact illustration: the production profile and judgement preference are related, but the relation is target-relative rather than automatic.

DATA AND CODE AVAILABILITY

The analysis code and derived files are in a public repository, withheld here for double-blind review and available to the editors on request; a de-anonymized link will accompany the camera-ready version. The analysis scripts fetch public datasets to temporary locations and validate downloaded files by their MD5 (message-digest 5) checksums.

File	MD5
BNC ₂₀₁₄ CSV	1c041dbcb635855f3d8f4be9f3e1fced
DAIS item CSV	46af310f1633b8784f897e4482faed84
DAIS cleaned-judgement ZIP	e7b3ac6f2dd8e85bc93e6e20fc0f6ec1

Derived files and the source-verification note are stored in the repository; the README gives their paths. The principal reproduction commands are:

```
Rscript analysis/08_dais_acceptability_bridge.R
Rscript analysis/09_bnc2014_metadata_scope.R
Rscript analysis/10_bnc2014_paired_transport_cv.R
python3 analysis/06_denominator_and_figure_candidates.py
make
```

The cross-validation and bootstrap seed is 20260625. Session information is written to a derived text file. Raw DAIS files aren't committed.

ACKNOWLEDGEMENTS

OpenAI Codex and Anthropic Claude assisted with drafting, coding, and review. They weren't used to create or manipulate submitted figures, images, or research data. All analysis decisions, source checks, and final text remain the author's responsibility.

REFERENCES

- Baayen, R. H., & Shafaei-Bajestan, E. (2025). *languageR: Data sets and functions with Analyzing Linguistic Data* [R package version 1.6]. Retrieved June 23, 2026, from <https://cran.r-project.org/package=languageR>
- Bresnan, J., Cueni, A., Nikitina, T., & Baayen, R. H. (2007). Predicting the dative alternation. In G. Bouma, I. Krämer & J. Zwarts (Eds.), *Cognitive foundations of interpretation* (pp. 69–94). Royal Netherlands Academy of Arts; Sciences.
- Bresnan, J., & Ford, M. (2010). Predicting syntax: Processing dative constructions in American and Australian varieties of English. *Language*, 86(1), 168–213.
- Bresnan, J., & Hay, J. (2008). Gradient grammar: An effect of animacy on the syntax of give in New Zealand and American English. *Lingua*, 118(2), 245–259.
- Charniak, E., Blaheta, D., Ge, N., Hall, K., Hale, J., & Johnson, M. (2000). *BLLIP 1987–89 WSJ corpus release 1* [LDC Catalog No. LDC2000T43]. Philadelphia, Linguistic Data Consortium. <https://doi.org/10.35111/fwew-da58>
- Engel, A., Andries, E., Rosseel, L., Szmrecsanyi, B., & Van de Velde, F. (2025). Constraints and lexical conditioning in the dative alternation: A cross-linguistic analysis of English and Dutch. *Languages in Contrast*, 25(2), 263–288. <https://doi.org/10.1075/lic.00052.eng>
- Godfrey, J. J., & Holliman, E. (1993). *Switchboard-1 Release 2* [LDC Catalog No. LDC97S62]. Philadelphia, Linguistic Data Consortium. <https://doi.org/10.35111/sw3h-rw02>
- Goodman, N. (1955). *Fact, fiction, and forecast*. Harvard University Press.
- Hawkins, R., Yamakoshi, T., Griffiths, T., & Goldberg, A. (2020a). *Dative Alternation and Information Structure (DAIS) judgements* [GitHub repository, commit 16ef145; CC BY 4.0]. Retrieved June 25, 2026, from https://github.com/taka-yamakoshi/neural_constructions
- Hawkins, R., Yamakoshi, T., Griffiths, T., & Goldberg, A. (2020b). Investigating representations of verb bias in neural language models. *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 4653–4663. <https://doi.org/10.18653/v1/2020.emnlp-main.376>
- Jenset, G., & McGillivray, B. (2018). *Data from “The dative alternation revisited: Fresh insights from contemporary British spoken data”*. figshare. <https://doi.org/10.6084/m9.figshare.7353164.v6>
- Jenset, G. B., & McGillivray, B. (2019). Data from “The dative alternation revisited: Fresh insights from contemporary British spoken data”. *Journal of Open Humanities Data*, 5(1). <https://doi.org/10.5334/johd.11>
- Kendall, T., Bresnan, J., & Van Herk, G. (2011). The dative alternation in African American English: Researching syntactic variation and change across sociolinguistic datasets. *Corpus Linguistics and Linguistic Theory*, 7(2), 229–244. <https://doi.org/10.1515/cllt.2011.011>
- Labov, W. (1972). *Sociolinguistic patterns*. University of Pennsylvania Press.
- Marcus, M. P., Santorini, B., Marcinkiewicz, M. A., & Taylor, A. (1999). *Treebank-3* [LDC Catalog No. LDC99T42]. Philadelphia, Linguistic Data Consortium. <https://doi.org/10.35111/gqix-j780>
- Röthlisberger, M., Grafmiller, J., & Szmrecsanyi, B. (2017). Cognitive indigenization effects in the English dative alternation. *Cognitive Linguistics*, 28(4), 673–710. <https://doi.org/10.1515/cog-2016-0051>

- Van Calster, B., Nieboer, D., Vergouwe, Y., De Cock, B., Pencina, M. J., & Steyerberg, E. W. (2016). A calibration hierarchy for risk models was defined: From utopia to empirical data. *Journal of Clinical Epidemiology*, 74, 167–176. <https://doi.org/10.1016/j.jclinepi.2015.12.005>
- Weinreich, U., Labov, W., & Herzog, M. I. (1968). Empirical foundations for a theory of language change. In W. P. Lehmann & Y. Malkiel (Eds.), *Directions for historical linguistics* (pp. 95–188). University of Texas Press.

SUPPLEMENTARY MATERIAL: THE DAIS ACCEPTABILITY BRIDGE

This bridge is supplementary to the external-validation result. It records what happens when the validated production model meets an acceptability channel, and it’s reported here for completeness rather than as part of the paper’s main claim.

Acceptability evidence answers a different question from corpus production. The Dative Alternation and Information Structure (DAIS) dataset, introduced by Hawkins et al. (2020b), contains 50,000 human judgements over 5,000 dative sentence pairs and systematically varies verb, definiteness, and argument length. The repository carries a CC BY 4.0 licence, and the dataset authors have given explicit permission to reuse the judgements, so DAIS can enter as a compact bridge (Hawkins et al., 2020a).

The bridge keeps the targets separate. I score only the 150 DAIS items whose verbs overlap the six-verb production core. The same spoken languageR production model that was transported to BNC2014 predicts an NP recipient probability for each constructed DAIS pair; the DAIS item mean gives the corresponding human double-object (DO) preference, preference for the same NP-recipient form, on a 0–1 scale. As a bridge analysis, it asks whether a production probability profile and a controlled preference profile point in the same direction.

The cleaned DAIS responses use integer values from 0 to 100 and are scaled here by division by 100. The shared-verb subset contains 1,525 judgement rows from 923 participants over 150 constructed items. Item support ranges from 6 to 14 judgements, with a median of 10; the mean item-level standard error is 0.082 on the 0–1 response scale.

Verb	Prod. NP prob.	DAIS DO pref.	Prod. - DAIS
<i>give</i>	0.408	0.449	-0.041
<i>lend</i>	0.204	0.426	-0.222
<i>offer</i>	0.150	0.468	-0.319
<i>sell</i>	0.231	0.351	-0.120
<i>send</i>	0.196	0.428	-0.232
<i>show</i>	0.506	0.516	-0.009

Table 4: DAIS bridge for the six shared verbs. The difference column is production probability minus DAIS DO preference.

The two channels line up moderately, while remaining distinct. With *something* treated as a pronominal theme, Pearson’s r is 0.667 and Spearman’s ρ is 0.685 over the 150 shared-verb items. The mean production NP probability is 0.283, while mean DAIS DO preference is 0.440; the mean paired difference is -0.157 , with bootstrap interval $[-0.198, -0.115]$. Treating *something* as nonpronominal gives the same interpretation ($r = 0.645$, $\rho = 0.675$, mean production probability 0.343). The verb means show a general calibration difference: mean production probability is below mean DAIS DO preference for every verb, not only for *offer*.

The item and participant checks show why the raw correlation shouldn’t be overread. At the item level, a marginal calibration regression has a positive slope, but once recipient condition, theme type, and verb are represented directly, the production-score slope is essentially zero (0.006, standard error 0.078).

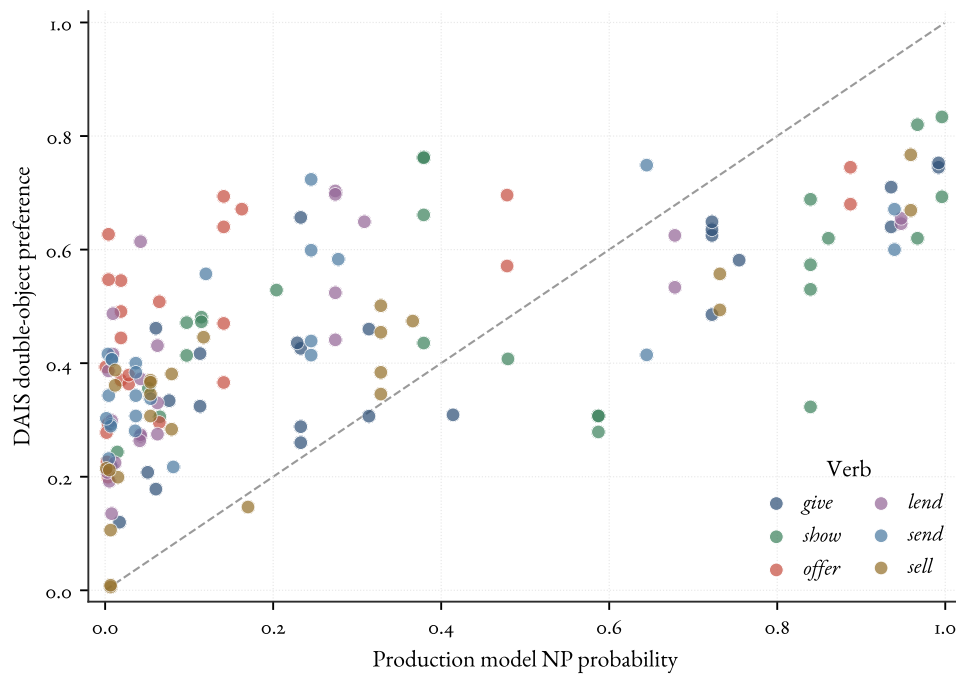


Figure 4: DAIS production/preference bridge for the six shared verbs. Each point is a constructed DAIS item. The horizontal axis is the spoken languageR model's NP recipient probability; the vertical axis is DAIS DO preference. The diagonal marks equality between the two channels; points are coloured by verb.

At the judgement level, a linear mixed model with Gaussian residuals uses fixed effects for production score, recipient condition, theme type, and verb, with crossed random intercepts for participant and constructed item. The incremental production-score coefficient is 0.017 (standard error 0.076; approximate interval $[-0.132, 0.165]$). DAIS preference and production probability show moderate marginal rank correspondence, much of it attributable to shared sensitivity to the manipulated predictors; the production score adds little once those predictors are represented directly.

The useful cases are calibrated divergences, not isolated raw gaps. After a marginal production–preference calibration, the verb-level residual is 0.077 for *offer* (bootstrap interval 0.027 to 0.129) and -0.070 for *sell* (interval -0.119 to -0.027). For example, *Juan offered the woman a payment* has predicted NP probability 0.141 but DAIS DO preference 0.694. The point is channel separation: judgement preference can occupy a different region of the evidence space from corpus production probability, and the difference isn’t reducible to a single verb anecdote.